The documentation and process conversion measures necessary to comply with this revision shall be completed by 31 October 2018.

INCH-POUND

MIL-PRF-19500/564J
w/AMENDMENT 1
31 July 2018
SUPERSEDING
MIL-PRF-19500/564J
1 November 2014

PERFORMANCE SPECIFICATION SHEET

* TRANSISTOR, FIELD EFFECT, P-CHANNEL, SILICON,
TYPES 2N6849, AND 2N6851,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a P-channel, enhancement-mode, MOSFET, power transistor for use in particular power-switching applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type.

1.2 Package outlines. The device package outlines are as follows: TO-205AF (formerly TO-39) in accordance with figure 1, surface mount version, "U" suffix (leadless chip carrier, LCC), see figure 2 for dimensions. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: "A" version die in accordance with figure 3 and "B" version die in accordance with figure 4.

1.3 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Type | $P_T$ (1) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C | $R_{\theta JC}$ (2) | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $I_{D1}$ (3) (4) $T_C$ = +25°C | $I_{D2}$ (3) (4) $T_C$ = +100°C | $I_S$ | $I_{DM}$ (5) | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | V dc | V dc | V dc | A dc | A dc | A dc | A (pk) | °C |
| 2N6849, 2N6849U | 25 | 0.8 | 5.0 | -100 | -100 | ±20 | -6.5 | -4.1 | -6.5 | -25 | -55 to +150 |
| 2N6851, 2N6851U | 25 | 0.8 | 5.0 | -200 | -200 | ±20 | -4.0 | -2.4 | -4.0 | -16 | -55 to +150 |

(1) Derate linearly by 0.2 W/°C for $T_C$ > +25°C.
(2) See figure 5, thermal impedance curves.
(3) The following formula derives the maximum theoretical $I_D$ spec. $I_D$ is limited by package and device construction.

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(4) See figure 6, maximum drain current graphs.
(5) $I_{DM}$ = 4 x $I_{D1}$ as calculated in note 2.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Unless otherwise specified, $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 $I_D$ = -1 mA dc | $V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ $I_D$ = -0.25 mA | | Max $I_{DSS1}$ $V_{GS}$ = 0 $V_{DS}$ = 80 percent of rated $V_{DS}$ | Max $r_{DS(on)1}$ (1) $V_{GS}$ = -10 V dc $I_D = I_{D2}$ | |
|---|---|---|---|---|---|---|
| | | | | | $T_J$ = +25°C | $T_J$ = +150°C |
| | V dc | V dc Min | Max | µA dc | ohm | ohm |
| 2N6849, 2N6849U | -100 | -2.0 | -4.0 | -25 | 0.30 | 0.60 |
| 2N6851, 2N6851U | -200 | -2.0 | -4.0 | -25 | 0.80 | 1.68 |

(1) Pulsed, (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.5.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

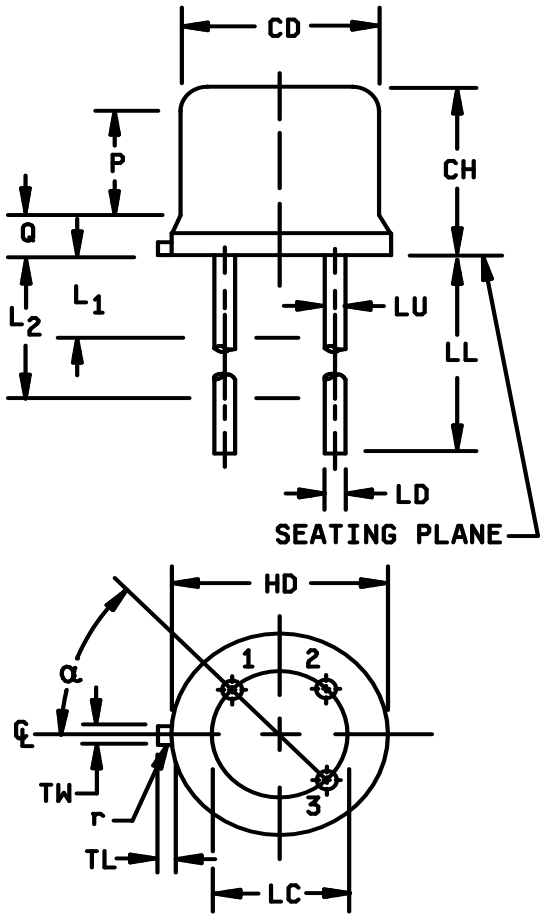
1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "6849" and "6851".

1.5.4 Suffix letters. The suffix letter "U" indicates that the transistor is a surface mount type package.

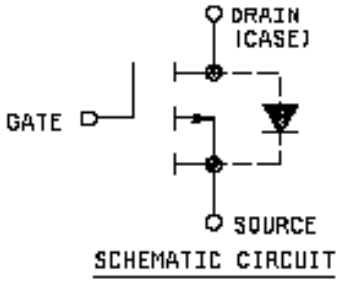
1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.



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| Ltr | Dimensions | | | | |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | Notes |
| | Min | Max | Min | Max | |
| CD | .315 | .355 | 8.01 | 9.01 | |
| CH | .160 | .180 | 4.07 | 4.57 | |
| HD | .340 | .370 | 8.64 | 9.39 | |
| LC | .200 TP | | 5.08 TP | | 6 |
| LD | .016 | .019 | 0.41 | 0.48 | 7, 8 |
| LL | .500 | .750 | 12.7 | 19.05 | 7, 8 |
| LU | .016 | .019 | 0.41 | 0.48 | 7, 8 |
| $L_1$ | | .050 | | 1.27 | 7, 8 |
| $L_2$ | .250 | | 6.35 | | 7, 8 |
| P | .100 | | 2.54 | | 5 |
| Q | | .050 | | 1.27 | 4 |
| r | | .010 | | 0.25 | 9 |
| TL | .029 | .045 | 0.74 | 1.14 | 3 |
| TW | .028 | .034 | 0.72 | 0.86 | 2 |
| α | 45° TP | | 45° TP | | 6 |

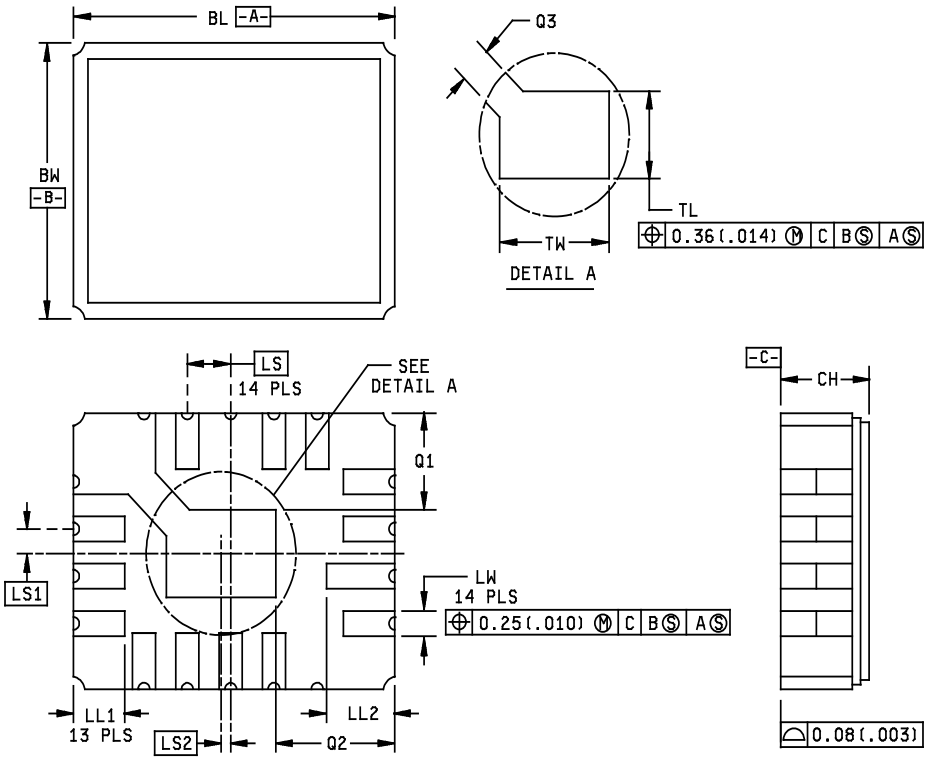


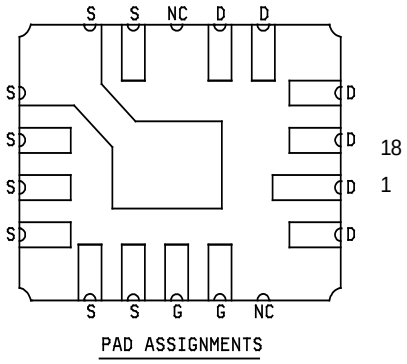

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Beyond radius (r) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
3. Dimension TL measured from maximum HD.
4. Outline in this zone is not controlled.
5. Dimension CD shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. LU applies between $L_1$ and $L_2$. LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
8. All three leads.
9. Radius (r) applies to both inside corners of tab.
10. Drain is electrically connected to the case.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = source, lead 2 = gate, lead 3 = drain.

* FIGURE 1. Physical dimensions for TO-205AF (2N6849 and 2N6851).



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| Dimensions | | | | |
|---|---|---|---|---|
| Sym. | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .345 | .360 | 8.77 | 9.14 |
| BW | .280 | .295 | 7.11 | 7.49 |
| CH | .095 | .115 | 2.41 | 2.92 |
| LL1 | .040 | .055 | 1.02 | 1.40 |
| LL2 | .055 | .065 | 1.40 | 1.65 |
| LS | .050 BSC | | 1.27 BSC | |
| LS1 | .025 BSC | | 0.635 BSC | |
| LS2 | .008 BSC | | 0.203 BSC | |
| LW | .020 | .030 | 0.51 | 0.76 |
| Q1 | .105 REF | | 2.67 REF | |
| Q2 | .120 REF | | 3.05 REF | |
| Q3 | .045 | .055 | 1.14 | 1.40 |
| TL | .070 | .080 | 1.78 | 2.03 |
| TW | .120 | .130 | 3.05 | 3.30 |

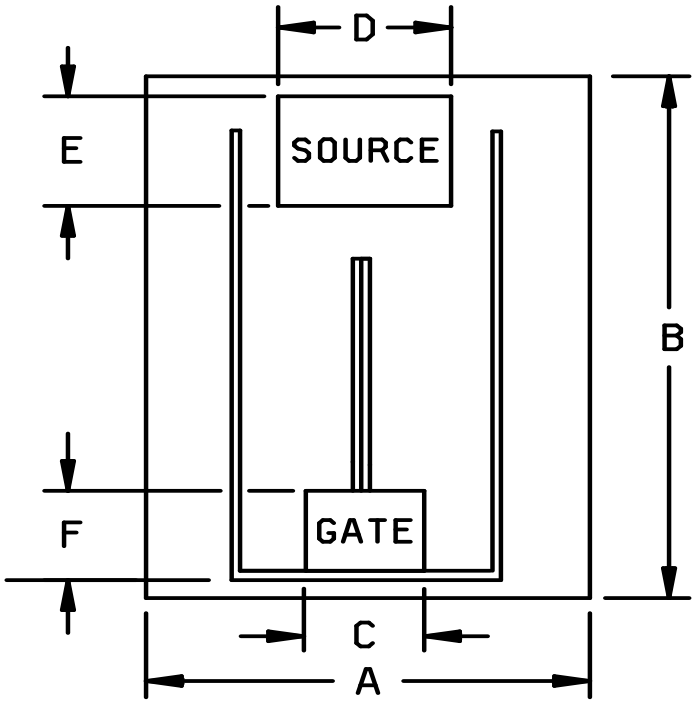
NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to $\phi$x symbology.

FIGURE 2. Physical dimensions for LCC (2N6849U and 2N6851U).



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2N6849 and 2N6851



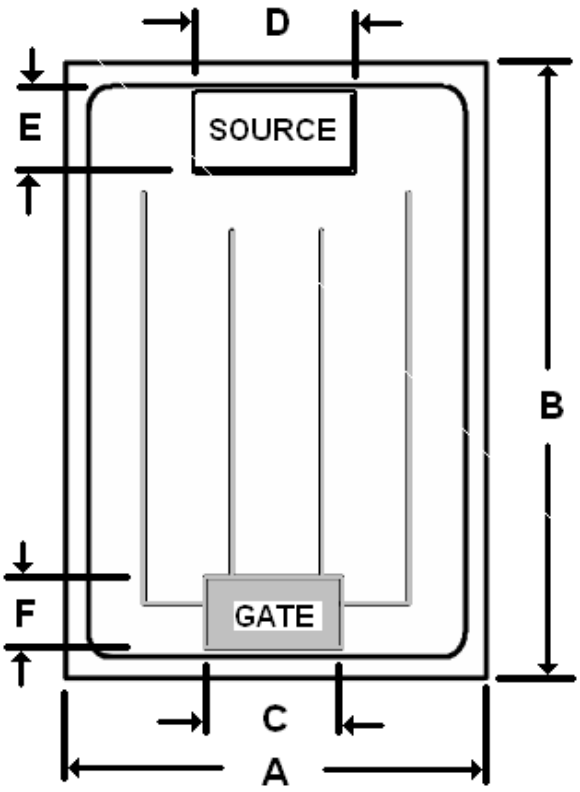

| Dimensions | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Ltr | 2N6849 | | | | 2N6851 | | | |
| | Inches | | Millimeters | | Inches | | Millimeters | |
| | Min | Max | Min | Max | Min | Max | Min | Max |
| A | .106 | .122 | 2.69 | 3.10 | .108 | .124 | 2.74 | 3.15 |
| B | .172 | .188 | 4.37 | 4.78 | .173 | .189 | 4.39 | 4.80 |
| C | .021 | .029 | 0.53 | 0.74 | .022 | .030 | 0.56 | 0.76 |
| D | .035 | .043 | 0.89 | 1.09 | .030 | .038 | 0.76 | 0.97 |
| E | .028 | .036 | 0.71 | 0.91 | .021 | .029 | 0.53 | 0.74 |
| F | .014 | .022 | 0.36 | 0.56 | .012 | .020 | 0.30 | 0.51 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and comprise the drain. The top metal is aluminum.
4. Die thickness is .0187 inch (0.475 mm), the tolerance is ±.0050 inch (±0.13 mm).

FIGURE 3. JANHCA and JANKCA die dimensions.



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| Dimensions | | | | |
|---|---|---|---|---|
| Ltr | 2N6849 | | | |
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .116 | .120 | 2.95 | 3.05 |
| B | .181 | .185 | 4.60 | 4.70 |
| C | .032 | .034 | 0.81 | 0.86 |
| D | .035 | .037 | 0.89 | 0.94 |
| E | .024 | .026 | 0.61 | 0.66 |
| F | .018 | .019 | 0.46 | 0.48 |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are: The back metals are chromium, nickel, and silver and comprise the drain. The top metal is aluminum.
4. Die thickness is .015 inch (0.381 mm), the tolerance is ±.0010 inch (±0.025 mm).

FIGURE 4. JANHCB die dimensions (2N6849 only).



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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

$I_{AS}$ .....Rated avalanche current, nonrepetitive.
nC .....nano Coulomb.

3.4 Interface and physical dimensions. The Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (TO-205AF), 2 (LCC), and 3 and 4 (JANHC/JANKC die) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.



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* 3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R ≤ 100 kΩ, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and table II herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-PRF-19500.



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* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| | Screen (see table E-IV of MIL-PRF-19500) (1) (2) | Measurement | |
|---|---|---|---|
| | | JANS level | JANTX and JANTXV levels |
| | (3) | Gate stress test (see 4.3.2) | Gate stress test (see 4.3.2). |
| | (3) | Method 3470 of MIL-STD-750 (see 4.3.3), optional | Method 3470 of MIL-STD-750 (see 4.3.3), optional |
| | (3) 3c | Method 3161 of MIL-STD-750 (see 4.3.4) | Method 3161 of MIL-STD-750 (see 4.3.4) |
| * | 9 | Subgroup 2 of table 1 herein. | Subgroup 2 of table 1 herein. |
| | 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| * | 11 | Subgroup 2 of table I herein:<br><br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 µA dc or ±100 percent of initial value, whichever is greater. | Subgroup 2 of table I herein: |
| | 12 | Method 1042 of MIL-STD-750, test condition A, t = 240 hours | Method 1042 of MIL-STD-750, test condition A |
| | 13 | Subgroups 2 and 3 of table I herein;<br><br>$\Delta I_{GSSF1}$ = ±20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ± 100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 µA dc or ± 100 percent of initial value, whichever is greater.<br><br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value. | Subgroup 2 of table I herein;<br><br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 µA dc or ±100 percent of initial value, whichever is greater.<br><br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(th)1}$ = ±20 percent of initial value. |

(1) At the end of the test program, $I_{GSSF1}$, $I_{GSSR1}$, and $I_{DSS1}$ are measured.
(2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.



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4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100-percent probed in accordance with table I, subgroup 2.

4.3.2 Gate stress test. Apply $V_{GS}$ = ±30 V minimum for t = 250 µs minimum.

4.3.3 Unclamped inductive switching.

a. Peak current ($I_D$) ..... Rated $I_{D1}$.

b. Peak gate voltage ($V_{GS}$) ..... -10 V dc.

c. Gate to source resistor ($R_{GS}$)..... 25 Ω ≤ $R_{GS}$ ≤ 200 Ω.

d. Initial case temperature ($T_C$) ..... +25°C +10°C, -5°C.

e. Inductance (L)..... 100 µH ±10 percent.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Pulse repetition rate..... None.

4.3.4 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, (and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 µs max. See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein.

* 4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1051 | Test condition G. |
| B3 | 2075 | See 3.4.2 herein. |
| B3 | 2077 | Scanning electron microscope (SEM) qualification may be performed anytime prior to lot formation. |
| B4 | 1042 | Test condition D. The heating cycle shall be 1 minute minimum. |
| B5 | 1042 | Accelerated steady-state operation life; test condition A; $V_{DS}$ = rated $T_A$ = +175°C, t = 120 hours. Read and record $V_{(BR)DSS}$ (pre and post) at 1 mA = $I_D$. Read and record $I_{DSS1}$ (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10 percent and $I_{DSS1}$ shall not exceed 25 µA.<br><br>Accelerated steady-state gate stress; condition B, $V_{GS}$ = rated, $T_A$ +175°C, t = 24 hours. |
| B5 | 2037 | Bond strength; test condition D. |



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* 4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B2 | 1051 | Test condition G, 25 cycles. |
| B3 | 1042 | Test condition D. The heating cycle shall be 1 minute minimum. |
| B4 | 2075 | See 3.4.2. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; weight = 8 ounces, 3 arcs of 90 degrees (Not required for LCC). |
| C5 | 3161 | See 4.5.2, $R_{\theta JC(max)}$ = 5.0°C/W. |
| C6 | 1042 | Test condition D. The heating cycle shall be 1 minute minimum. |

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. The thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$ (and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See table E-IX of MIL-PRF-19500, group E, subgroup 4.



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* TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3161 | See 4.3.4 | $Z_{\theta JC}$ | | | °C/W |
| Breakdown voltage, drain to source | 3407 | Bias condition C, $V_{GS}$ = 0V, $I_D$ = -1 mA dc | $V_{(BR)DSS}$ | | | |
| 2N6849, 2N6849U | | | | -100 | | V dc |
| 2N6851, 2N6851U | | | | -200 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = -0.25 mA | $V_{GS(th)1}$ | -2.0 | -4.0 | V dc |
| Gate current | 3411 | Bias condition C, $V_{GS}$ = +20 V dc $V_{DS}$ = 0 | $I_{GSSF1}$ | | +100 | nA dc |
| Gate current | 3411 | Bias condition C, $V_{GS}$ = -20 V dc $V_{DS}$ = 0 | $I_{GSSR1}$ | | -100 | nA dc |
| Drain current | 3413 | Bias condition C, $V_{GS}$ = 0, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS1}$ | | -25 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = -10 V dc, condition A, pulsed (see 4.5.1), $I_D$ = rated $I_{D2}$ (see 1.3) | $r_{DS(on)1}$ | | | |
| 2N6849, 2N6849U | | | | | 0.30 | Ω |
| 2N6851, 2N6851U | | | | | 0.80 | Ω |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = -10 V dc, condition A, pulsed (see 4.5.1), $I_D$ = rated $I_{D1}$, (see 1.3) | $r_{DS(on)2}$ | | | |
| 2N6849, 2N6849U | | | | | 0.32 | Ω |
| 2N6851, 2N6851U | | | | | 0.83 | Ω |
| Forward voltage | 4011 | $V_{GS}$ = 0, $I_D$ = rated $I_{D1}$, condition A | $V_{SD}$ | | | V |
| 2N6849, 2N6849U | | | | | -4.3 | V |
| 2N6851, 2N6851U | | | | | -5.6 | V |

See footnote at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C = T_J = +125°C$ | | | | |
| Gate current | 3411 | Bias condition C, $V_{GS} = \pm 20$ V dc, $V_{DS} = 0$ | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | Bias condition C, $V_{GS} = 0$, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS2}$ | | -0.25 | mA dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D = -.25$ mA | $V_{GS(th)2}$ | -1.0 | | V dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = -10 V dc, condition A, pulsed (see 4.5.1), $I_D$ = rated $I_{D2}$ | $r_{DS(on)3}$ | | | |
| 2N6849, 2N6849U | | | | | 0.54 | Ω |
| 2N6851, 2N6851U | | | | | 1.60 | Ω |
| Low temperature operation: | | $T_C = T_J = -55°C$ | | | | |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D = -.25$ mA | $V_{GS(th)3}$ | | -5.0 | V dc |
| Subgroup 4 | | | | | | |
| Switching time test | 3472 | $I_D$ = rated $I_{D1}$; $V_{GS}$ = -10 V dc; gate drive impedance = 7.5 Ω; | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | | |
| 2N6849, 2N6849U | | $V_{DD}$ = -40 V dc | | | 60 | ns |
| 2N6851, 2N6851U | | $V_{DD}$ = -75 V dc | | | 50 | ns |
| Rise time | | | $t_r$ | | | |
| 2N6849, 2N6849U | | $V_{DD}$ = -40 V dc | | | 140 | ns |
| 2N6851, 2N6851U | | $V_{DD}$ = -75 V dc | | | 100 | ns |
| Turn-off delay time | | | $t_{d(off)}$ | | | |
| 2N6849, 2N6849U | | $V_{DD}$ = -40 V dc | | | 140 | ns |
| 2N6851, 2N6851U | | $V_{DD}$ = -75 V dc | | | 80 | ns |
| Fall time | | | $t_f$ | | | |
| 2N6849, 2N6849U | | $V_{DD}$ = -40 V dc | | | 140 | ns |
| 2N6851, 2N6851U | | $V_{DD}$ = -75 V dc | | | 80 | ns |

See footnotes at end of table.



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* TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Condition | | Min | Max | |
| Subgroup 5 | | | | | | |
| Single pulse unclamped inductive switching 3/ | 3470 | See 4.3.3 | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Safe operating area test (high voltage) | 3474 | See figure 7<br>$t_p$ = 10 ms, $V_{DS}$ = 80 percent of rated $V_{DS}$, $V_{DS} \leq$200 V dc max. | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B | | | | |
| On-state gate charge | | | $Q_{g(on)}$ | | | |
| 2N6849, 2N6849U | | | | | 34.8 | nC |
| 2N6851, 2N6851U | | | | | 34.8 | nC |
| Gate to source charge | | | $Q_{gs}$ | | | |
| 2N6849, 2N6849U | | | | | 6.8 | nC |
| 2N6851, 2N6851U | | | | | 6.1 | nC |
| Gate to drain charge | | | $Q_{gd}$ | | | |
| 2N6849, 2N6849U | | | | | 23.1 | nC |
| 2N6851, 2N6851U | | | | | 20.5 | nC |
| * Reverse recovery time | 3473 | Condition A, $d_i/d_t \leq$ -100 A/µs, $V_{DD} \leq$ -50 V, | $t_{rr}$ | | | |
| 2N6849, 2N6849U | | $I_F$ = -6.5 A | | | 250 | ns |
| 2N6851, 2N6851U | | $I_F$ = -4.0 A | | | 400 | ns |

1/ For sampling plan, see MIL-PRF-19500.

* 2/ For end-point measurements, this test is required for the following subgroups (not intended for 4.3, screen 9, 11, or 13):
JANS, table E-VIA of MIL-PRF-19500, group B, subgroups 3 and 4; JAN, JANTX, and JANTXV, table E-VIB of MIL-PRF-19500, group B, subgroups 2 and 3; and table E-VII of MIL-PRF-19500, group C, subgroups 2 and 6, and table E-IX of MIL-PRF-19500, group E, subgroup 1.

3/ This test is optional if performed as a 100 percent screen.



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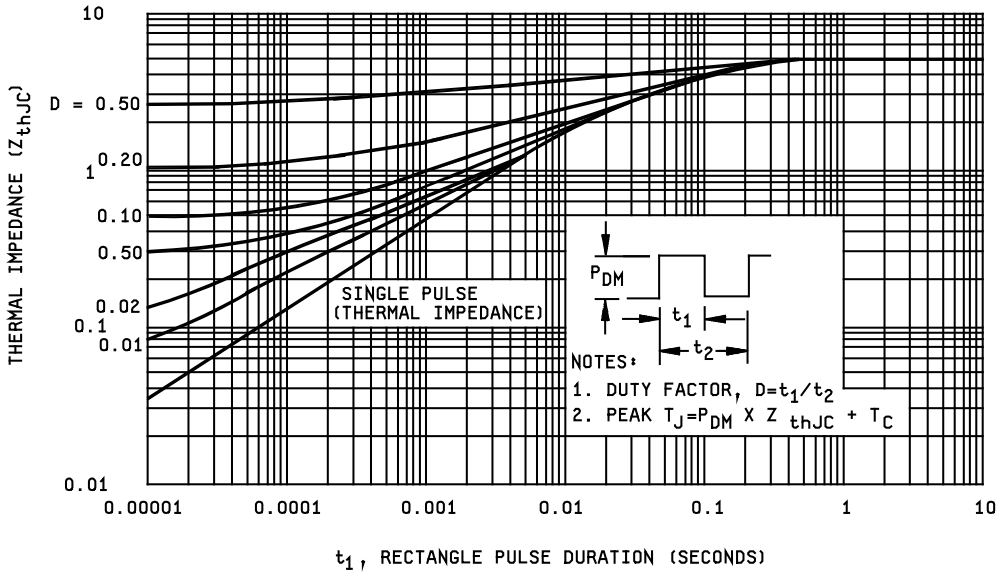
TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Qualification inspection |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycle | 1051 | Condition G, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 2 1/ | | | 45 devices<br>c = 0 |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 4 | | | Sample size<br>N/A |
| Thermal impedance curves | | See MIL-PRF-19500. | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 10 | | | |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | Test conditions shall be derived by the manufacturer | 22 devices<br>c = 0 |

1/ A separate sample for each test may be pulled.



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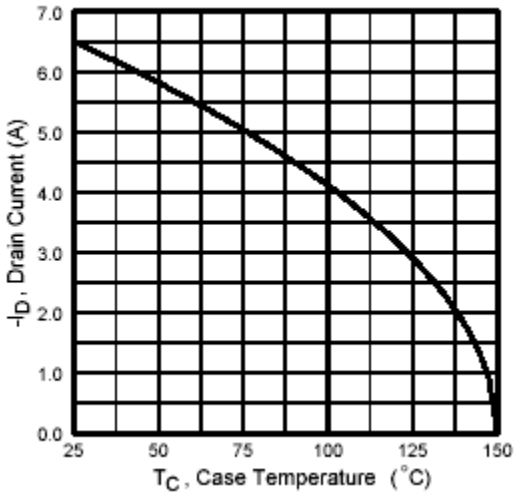



2N6849, 2N6849U, 2N6851, and 2N6851U

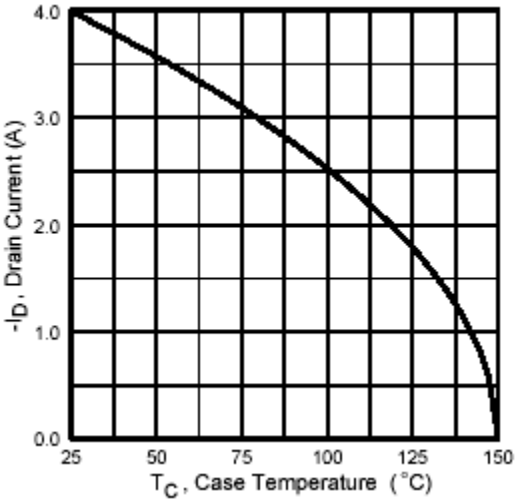
FIGURE 5. Normalized transient thermal impedance.



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2N6849, 2N6849U



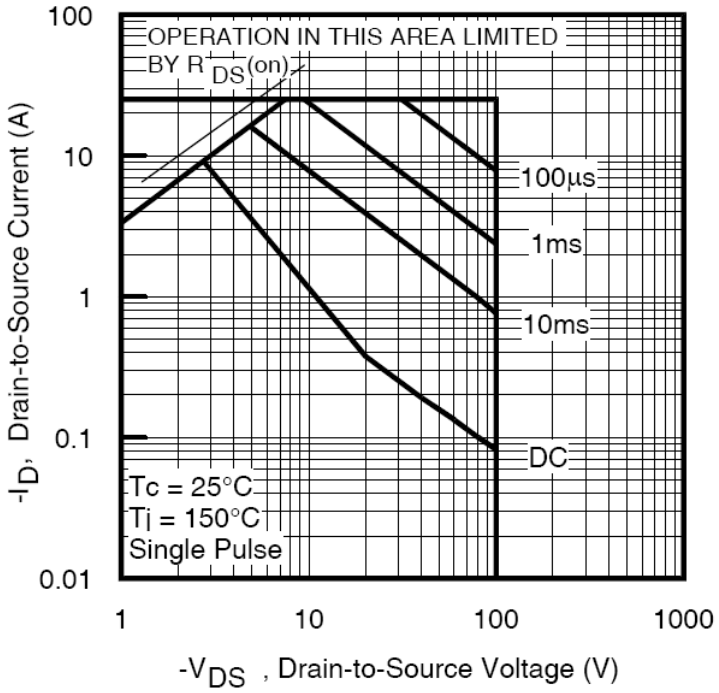

2N6851, 2N6851U

FIGURE 6. Maximum drain current versus case temperature graphs.



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ACTIVE REGION - 2N6849, 2N6849U




ACTIVE REGION - 2N6851, 2N6851U

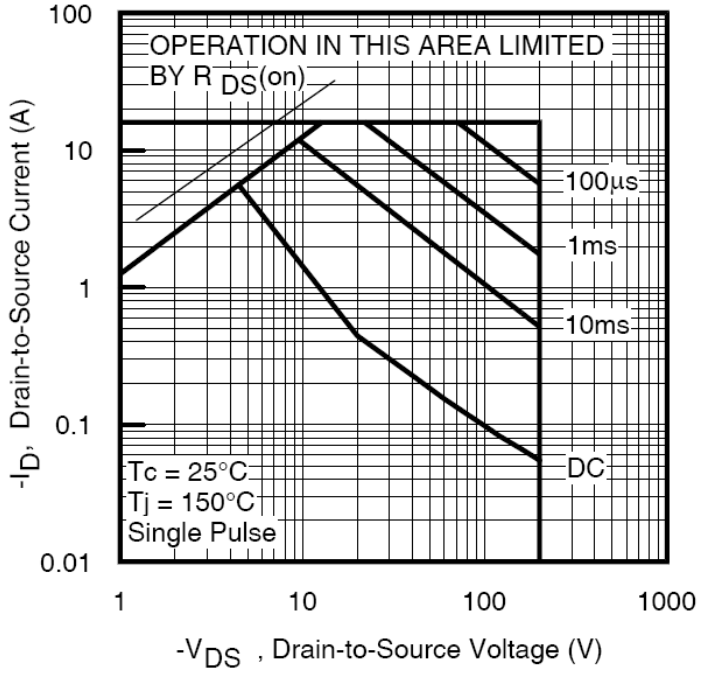



FIGURE 7. Maximum safe operating area.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see 1.5.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://qpldocs.dla.mil.

6.4 Cross-reference and complement list. Parts from this specification may be used to replace the following commercial Part or Identifying Number (PIN's). The term PIN is equivalent to the term part number which was previously used in this specification.

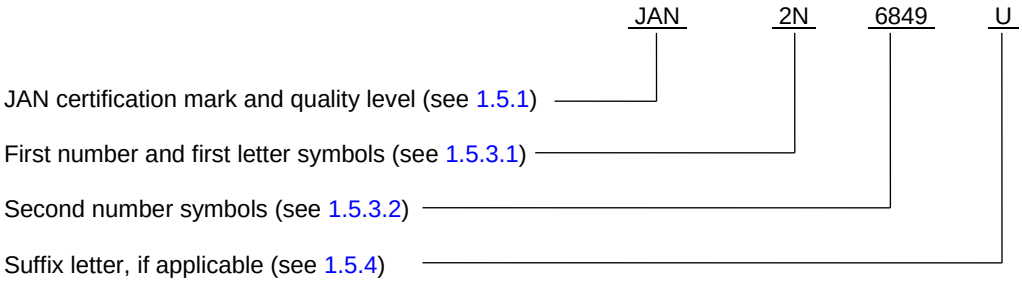
| Preferred types | Commercial types (1) |
|---|---|
| 2N6849 | IRFF9130, IRFF9131, IRFF9132, IRFF9133 |
| 2N6851 | IRFF9230, IRFF9231, IRFF9232, IRFF9233 |
| 2N6849U | IRFE9130, IRFE9131, IRFE9132, IRFE9133 |
| 2N6851U | IRFE9230, IRFE9231, IRFE9232, IRFE9233 |

(1) Complementary devices can be found on MIL-PRF-19500/557



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6.5 PIN construction example. The PINs for encapsulated devices are construction using the following form.




6.6 List of PINs.

6.6.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

| PINs for devices of the base quality level | PINs for devices of the "TX" quality level | PINs for devices of the "TXV" quality level | PINs for devices of the "S" quality level |
|---|---|---|---|
| JAN2N6849 | JANTX2N6849 | JANTXV2N6849 | JANS2N6849 |
| JAN2N6849U | JANTX2N6849U | JANTXV2N6849U | JANS2N6849U |
| JAN2N6851 | JANTX2N6851 | JANTXV2N6851 | JANS2N6851 |
| JAN2N6851U | JANTX2N6851U | JANTXV2N6851U | JANS2N6851U |

6.6.2 PINs for unencapsulated devices (die). The following is a list of possible PINs for unencapsulated devices available on this specification sheet.

| Quality level HC | Quality level KC |
|---|---|
| JANHCA2N6849 | JANKCA2N6849 |
| JANHCB2N6849 | |
| JANHCA2N6851 | JANKCA2N6851 |

6.7 Suppliers of JANC die. The qualified JANC suppliers with the applicable letter version (example JANHCA2N6849) will be identified on the QML.

| JANC ordering information | | |
|---|---|---|
| PIN | Manufacturer | |
| | 59993 | 43611 |
| 2N6849 | JANHCA2N6849<br>JANKCA2N6849 | JANHCB2N6849 |
| 2N6851 | JANHCA2N6851<br>JANKCA2N6851 | |



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* 6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

* 6.9 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Air Force - 85
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Review activities:
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